

Conditional Generation of Nanomaterials Using Diffusion Models for Enhanced Structure Prediction

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18.06.2025

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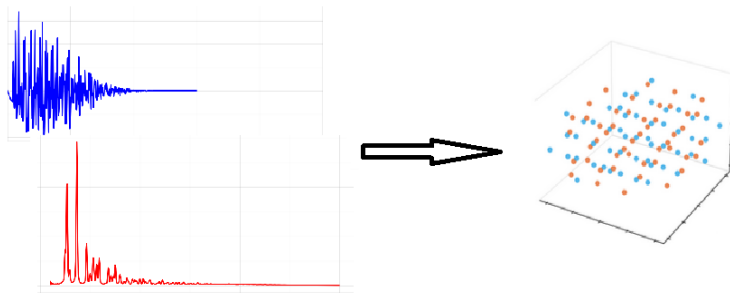
Motivation



Problem Statement

Inverse Design Challenge: Generate nanomaterial structures from experimental measurements

- Traditional: Structure $\rightarrow$ Properties $\rightarrow$ Synthesis
- **Our Goal:** Experimental Data $\rightarrow$ Structure Prediction
- Enables targeted materials discovery



Materials Discovery Bottleneck

Current Limitations:

- Experimental synthesis: Expensive with months of work
- Existing models limited: DiffCSP (52 atoms), MatterGen (20 atoms)
- Gap: Nanomaterials usually have more than 1000 atoms (CHILI)

Our Contribution: CNN-based diffusion for 100-atom nanomaterials with experimental conditioning + enhanced metrics for structural quality

Nanomaterials



Nanomaterials & Applications

Nanomaterials: Structures $< 100\text{nm}$ with unique properties

- **Applications:** Energy storage, catalysis, medicine, electronics
- **Unique properties:** High surface area, quantum effects, enhanced strength
- **Challenge:** $10^1 - 10^5$ atoms, expensive synthesis

Solution: Computational prediction before synthesis

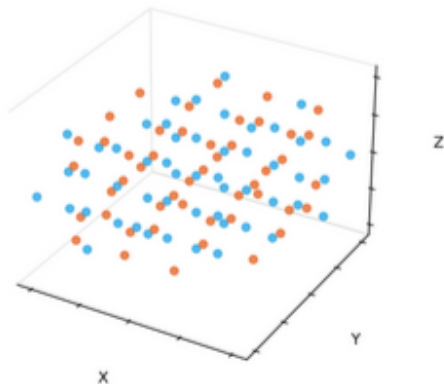
Structural Representation & Experimental Data

Data Representation:

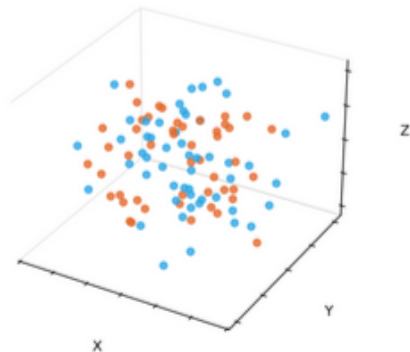
- **Coordinates:** Absolute (x,y,z) vs Fractional (unit cell fractions)
- **Experimental Signatures:** XRD (580 features) vs XPDF (6000 features)
- **Challenge:** Predict structure from experimental measurements

Predicting structure = Predicting properties = Designing materials for specific applications

Structural Representation

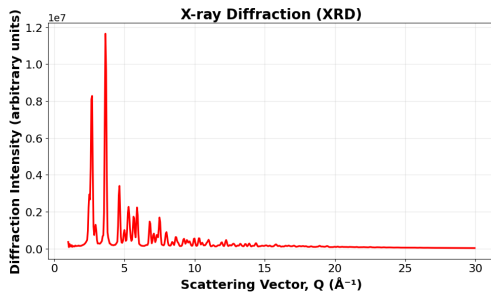


(a) Absolute Coordinates

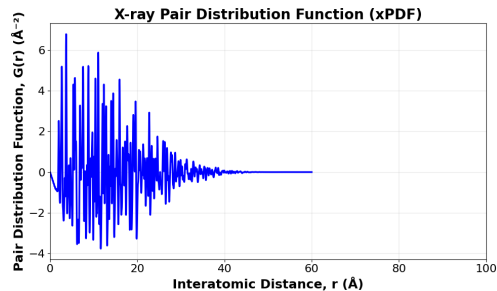


(b) Fractional Coordinates

Experimental Spectra



(a) XRD



(b) XPDF

Applications in Key Industries

Energy Storage & Conversion:

- Battery electrodes, supercapacitors, solar cells

Medicine & Biology:

- Drug delivery, MRI contrast agents, antimicrobial materials

Environmental Applications:

- Water treatment (TiO_2), carbon capture, pollution control

Agriculture:

- Crop bio-fortification, smart fertilizers

Structural Complexity Challenge

Crystallographic Complexity:

- **7 Crystal Systems:** Different lattice geometries
- **14 Bravais Lattices:** Primitive, body-centered, face-centered
- **32 Point Groups:** Rotational symmetries
- **230 Space Groups:** Complete symmetry description

Challenge: Enormous search space for algorithms to explore

Solution: Learn patterns from data rather than explicit constraints

Diffusion Models



Diffusion Process

Forward Process: Gradually add noise until pure Gaussian noise

Reverse Process: Model learns to predict and remove noise

- Sampling: Iteratively denoise from noise to data
- Advantage: High-quality, diverse generation

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t\mathbf{I}) \quad (1)$$

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \quad (2)$$

Hybrid Continuous-Discrete Diffusion

Key Innovation: Handle multiple data types in single framework

- **Coordinates:** Continuous diffusion (DDPM)
- **Atom types:** Discrete diffusion (D3PM)
- **Conditioning:** Direct experimental data integration

Advantage: Single model, shared conditioning, consistent generation

Diffusion Sampling Strategies

DDPM Sampling (Stochastic):

- Adds random noise at each step
- Higher sample diversity
- More exploration of chemical space

DDIM Sampling (Deterministic):

- No random component
- Faster generation (skip steps possible)
- Consistent results from same starting point

Our Choice: DDPM for better exploration of atomic arrangements

Conditioning Approaches

Classifier Guidance:

- Uses separate classifier $p_\phi(y|x_t, t)$
- Guides with gradients $\nabla_{x_t} \log p_\phi(y|x_t, t)$

Classifier-Free Guidance:

- Joint training: conditional + unconditional
- $\tilde{\epsilon}_\theta = (1 + w)\epsilon_\theta(x_t, c) - w\epsilon_\theta(x_t, \emptyset)$

Direct Conditioning (Our Approach):

- Concatenate experimental features directly
- Efficient: single forward pass
- Rich conditioning: XRD/XPDF provides sufficient guidance

Discrete Diffusion for Atom Types

Challenge: Cannot add Gaussian noise to categorical data

Solution - Transition Matrices:

$$Q_t = (1 - \beta_t)I + \beta_t M$$

- I : Identity matrix (no corruption)
- M : Target corruption pattern (uniform distribution)
- β_t : Noise schedule (same as continuous)

Process: Atoms gradually transition to masked state, then model predicts original types

Advantage: Same timestep t for both coordinates and atom types

Implementation



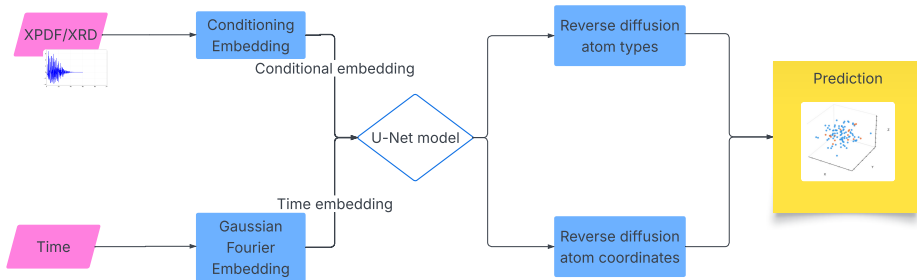
Novel CNN Architecture vs Traditional GNNs

CNN-Based Approach:

- U-Net with 10×10 grid representation (100 atoms)
- Channel progression: [64, 128, 256, 512]
- Dual outputs: coordinate noise + atom type logits
- **Advantage:** Leverages decades of CNN optimization

Why CNN? More accessible than specialized geometric frameworks

Architecture



Training & Experimental Conditioning

Combined Loss Function:

$$L = L_{\text{continuous}} + \lambda L_{\text{discrete}}$$

Training Details:

- Adam optimizer with EMA stabilization
- Exponential learning rate decay ($\gamma = 0.9999$)
- Direct conditioning via experimental data concatenation

Conditioning: XRD (580) or XPDF (6000) features embedded and broadcast

Evaluation and Results



Baseline Model

Multi-Layer Perceptron (MLP) Baseline:

- **Architecture:** 3-layer MLP with residual blocks
- **Input:** XRD/XPDF experimental data
- **Output:** Direct prediction of coordinates + atom types
- **Training:** Same loss function as diffusion model

Comparison Purpose:

- Tests whether iterative diffusion offers advantages
- More efficient (single forward pass vs multiple steps)
- Lacks theoretical benefits of noise-to-data transformation

Evaluation Metrics

Problem: Existing metrics ignore rotation and atom ordering

Our Solution:

- **Kabsch Algorithm:** Optimal rotational alignment
- **Hungarian Algorithm:** Optimal atom-to-atom assignment
- **Result:** 59% improvement in meaningful distance metrics

Impact: More accurate evaluation for ALL structure generation methods

Alignment comparison

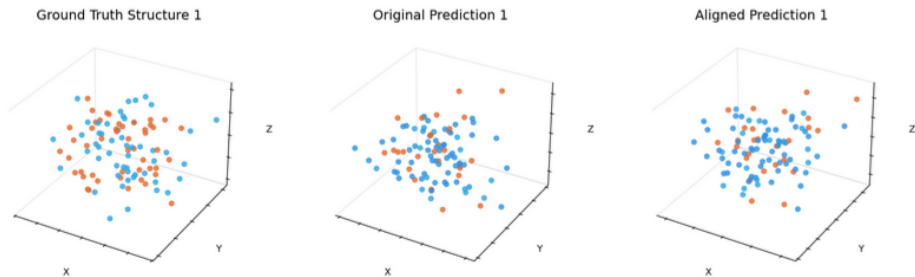


Figure: Alignment comparison

Key Results: Performance Comparison

Model	Atom Type Acc	MAE	Optimized MAE	Typed MAE
MLP Baseline	14.5%	1.27 Å	1.22 Å	17.66Å
Diffusion T100	69.57%	1.68 Å	0.63 Å	6.50 Å
Diffusion T1000	69.04%	1.80 Å	0.67 Å	6.62 Å

Table: Test metrics for fractional coordinates conditioned on XPDF

Key Achievements:

- $4.7\times$ improvement in atom type prediction
- Superior spatial prediction with optimized metrics
- Substantial improvement over direct regression

Conditioning & Coordinate System Comparison

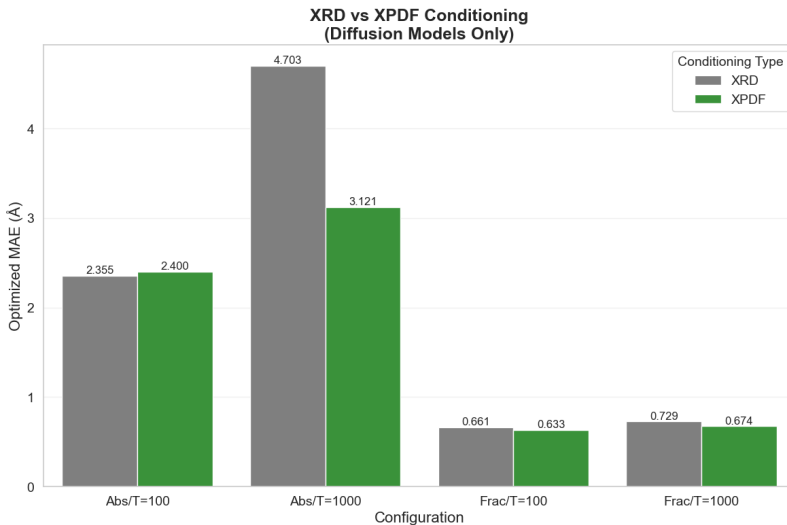
XPDF vs XRD Conditioning:

- XPDF performs similarly/better (6000 vs 580 features)
- Richer structural information enables better generation

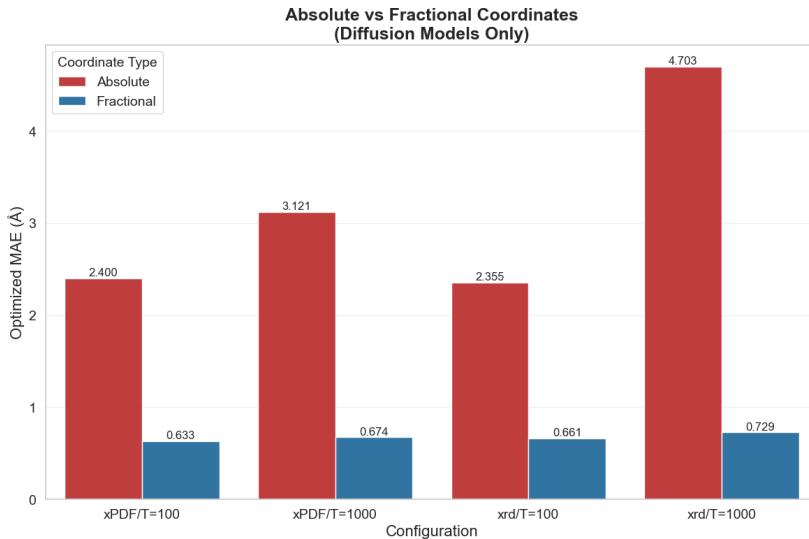
Fractional vs Absolute Coordinates:

- Fractional coordinates more suitable for diffusion
- Normalized $[0,1]$ range improves learning
- **Limitation:** Requires lattice parameters for reconstruction

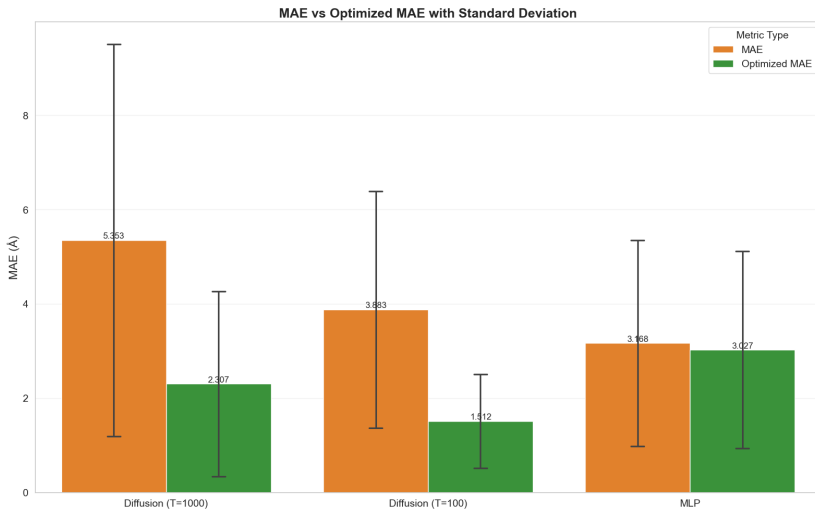
XPDF vs XRD comparison



Fractional vs Absolute Coordinates



Quality of Alignment



Qualitative Results

Generated Structure Quality:

- Reasonable atom types and spatial arrangements
- Improved alignment reveals true structural similarity
- Some challenges with atom clustering (future work)

Visualizations

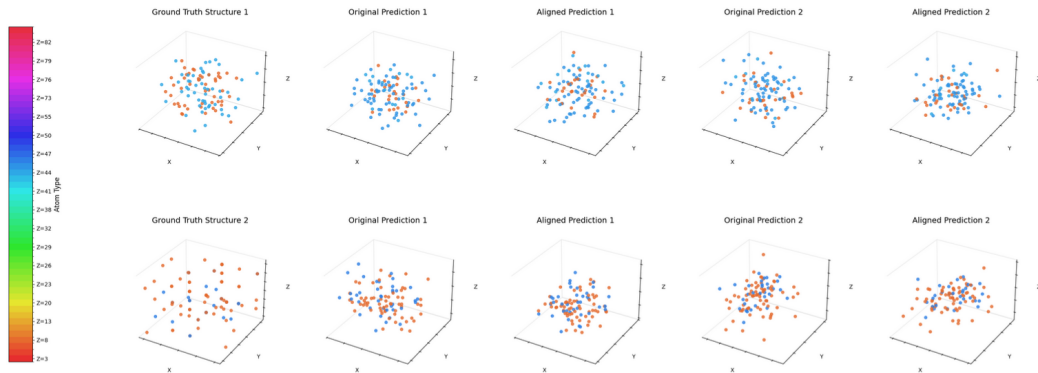


Figure: Visualization of generated structures

Discussion



Contributions & Context

Our Contributions:

- Novel CNN-based approach for materials generation
- Experimental conditioning for inverse design (XRD/XPDF)
- Improved evaluation methodology (Kabsch+Hungarian)
- Scaling to 100 atoms (vs 20-52 in existing work)

Recent Context: ADiT, MatterGen validate diffusion for materials

→ Our CNN approach provides alternative paradigm

Limitations & Future Work

Current Limitations:

- Sequential reshaping doesn't preserve spatial relationships
- Limited to 100 atoms (subset of larger structures)
- Mono-metallic oxide chemical scope
- Physical and chemical validity

Future Directions:

- Predict lattice parameters for fractional coordinates
- Spatial-aware grid representations
- Extension to larger, more diverse structures
- Multiple experimental conditioning sources

Conclusion



Conclusion

Key Achievements:

- Demonstrated CNNs can effectively generate nanomaterial structures
- 70% atom type accuracy ($4.7\times$ improvement over baselines)
- Novel experimental conditioning enables inverse design
- Improved evaluation methodology benefits entire field
- Foundation for accessible CNN-based materials generation

Thank You
for Your Attention

